Taehei Kim

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Google Scholar · Daejeon, South Korea

Research interests Mixed reality telepresence; 3D scene synthesis and generation; human-centered XR systems.

Professional Experience

2026.09– **Postdoctoral Researcher**, InnoCORE AI-Meta Scientist Program, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea.
Host: Prof. Tae-Kyun Kim.

2024.09– **Co-Founder**, BerryHi, Daejeon, South Korea.
Mixed reality telepresence systems; lead research project funded by RAPA and shipped *Parameter Room* on the Meta Quest Store.

Education

2021.03–2026.08 **Ph.D., Culture Technology**, KAIST, Daejeon, South Korea.
Advisor: Prof. Sung-Hee Lee.
Dissertation: *Dynamic Space Optimization and Adjoining for Transmitting User and Spatial Context in Mixed Reality Telepresence*.

2018.08–2021.02 **M.S., Culture Technology**, KAIST, Daejeon, South Korea.
Advisor: Prof. Sung-Hee Lee.
Thesis: *Quadruped Locomotion on Non-Rigid Terrain using Reinforcement Learning*.

2013.03–2018.08 **B.A. in Asian Studies** and **B.S. in Computer Science** (dual degree), Yonsei University, Seoul, South Korea.

Honors and Awards

2026.07 Best Poster Award, Korea Computer Graphics Society (KCGS).

2025.11 Excellence Award, Metaverse Developer Contest, Korea Radio Promotion Association (RAPA).

2024.10 Best Demo Honorable Mention, IEEE ISMAR.

2021.03–2026.08 Fully Funded Ph.D. Fellowship, KAIST.

2021.09 Tenacity Scholarship, KAIST.

2021.08 First Place, "Hello, World!: New Playground", Art Center NABI.

2018.08–2021.02 Fully Funded M.S. Fellowship, KAIST.

Publications

†first author *equal contribution

Peer-Reviewed Journal Articles

[J4] **Taehei Kim**†, J. Shin, H. Kim, H. Jang, J. Kang, and S.-H. Lee. "Voronoi Rooms: Dynamic Visibility Modulation of Overlapping Spaces for Telepresence." *ACM Transactions on Graphics (TOG)*, 45(2), 2026. doi:10.1145/3777900. Presented at ACM SIGGRAPH 2026.

[J3] J. Kang, **Taehei Kim**, H. Kim, and S.-H. Lee. "Real-Time Translation of Upper-Body Gestures to Virtual Avatars in Dissimilar Telepresence Environments." *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 31(10):8711–8725, 2025. doi:10.1109/TVCG.2025.3577156.

[J2] **Taehei Kim**†, H. Kim, J. Lee, and S.-H. Lee. "Evaluating User Perception toward Physics-Adapted Avatars in Remote Heterogeneous Spaces." *Computers & Graphics*, 128:104185, 2025.doi: 10.1016/j.cag.2025.104185

[J1] D. Yang, J. Kang, **Taehei Kim**, and S.-H. Lee. "Visual Guidance for User Placement in Avatar-Mediated Telepresence between Dissimilar Spaces." *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 30(12):7558–7570, 2024. doi: 10.1109/TVCG.2024.3354256

Peer-Reviewed Full Conference Papers

[C3] **Taehei Kim**†, H. Kim, J. Shin, H. Kim, and S.-H. Lee. "Neighborhood: Dynamic Spatial Alignment for Traversable Telepresence across Remote Spaces." *ACM Symposium on User Interface Software and Technology (UIST)*, 2026. (To Appear)

[C2] H. Kim, **Taehei Kim***, J. Shin*, H. Kim, and S.-H. Lee. "TranSpace: Progressive Anchoring for Metric-Consistent Scene Synthesis." *ACM International Conference on Multimedia (MM)*, 2026. doi:10.1145/3767308.3836316.

[C1] J. Kang, D. Yang, **Taehei Kim**, Y. Lee, and S.-H. Lee. "Real-Time Retargeting of Deictic Motion to Virtual Avatars for Augmented Reality Telepresence." *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 885–893, 2023. doi: 10.1109/ISMAR59233.2023.00104

Peer-Reviewed Short Papers, Posters, and Demonstrations

[D4] J. Shin, H. Kim, E. Lee, D. Shin, K. Lee, **Taehei Kim**, H. Kim, J. An, and S.-H. Lee. "Situated Embodied XR Agents via Spatial Reasoning and Prompting." *IEEE ISMAR Adjunct*, pp. 933–934, 2025. doi:10.1109/ISMAR-Adjunct68609.2025.00255.

[D3] **Taehei Kim**†, J. Shin, H. Kim, H. Jang, J. Kang, and S.-H. Lee. "Visibility Modulation of Aligned Spaces for Multi-User Telepresence." *IEEE ISMAR Adjunct*, pp. 626–627, 2024. **Best Demo Honorable Mention.**

[D2] H. Jang*, **Taehei Kim***, S. Oh, J. Lee, S. Lee, and S. H. Yoon. "Sense of Embodiment Inducement for People with Reduced Lower-Body Mobility and Sensations with Partial-Visuomotor Stimulation." *ACM SIGGRAPH Emerging Technologies*, pp. 1–2, 2022.

[D1] H.-I. Kim, **Taehei Kim**, E. Song, S. Y. Oh, D. Kim, and W. Woo. "Multi-Scale Mixed Reality Collaboration for Digital Twin." *IEEE ISMAR Adjunct*, pp. 435–436, 2021.

Preprints and Manuscripts under Review

[P2] One paper currently under review, 2026.

[P1] **Taehei Kim**† and S.-H. Lee. "Quadruped Locomotion on Non-Rigid Terrain using Reinforcement Learning." *arXiv:2107.02955*, 2021.

Grants and Funded Research Projects

2026.06– KEP Startup Fund — *Real-Space Overlay Scene Generation for VR Crowdsourcing Data Collection*. USD 37,000. Role: Berryhi internal director.

2024.05–2025.12 Korea Radio Promotion Association (RAPA) — *Mixed Reality Telepresence System*. Role: Student project lead.

2024.08–2024.09 Daejeon Art Center Joint New Technology Performance. USD 11,000. Role: Berryhi lead director.

2022.03–2024.02 National Research Foundation of Korea — *Mixed Reality Scenes in Heterogeneous Spaces*. Role: Student project lead.

Demonstrations, Exhibitions, and Deployed Systems

2025.11 *Parameter Room*, Mixed reality telepresence application, Meta Quest Store (BerryHi).

2024.10 *Visibility Modulation of Aligned Spaces for Multi-User Telepresence*, IEEE ISMAR 2024 Demonstrations, Seattle, USA. Best Demo Honorable Mention.

2024.09 *X-Space*, Daejeon Art Center, Daejeon, South Korea.

2022.07 *Sense of Embodiment Inducement with Partial-Visuomotor Stimulation*, ACM SIGGRAPH 2022 Emerging Technologies, Vancouver, Canada.

Teaching Experience

Guest Lectures and Program Designs

2021.08–2023.07 Metaverse Instructor. Designed and delivered XR content-creation curricula for public and corporate audiences. Clients: Dangjin Cultural Foundation, Daejeon Culture and Arts Foundation, Hanwha Galleria, Daejeon Hannam University, 3PROTV and more.

Courses Prepared to Teach

Computer Graphics; Virtual and Augmented Reality; Human–Computer Interaction; Generative Models and Agents for 3D Content Generation.

Mentoring

2024.03– Mentored M.S. students (Hyeshim Kim, Jihun Shin, Hyeonjin Kim) at KAIST LAVA Lab and in collaborating laboratories (Taeyeon Kim). Mentorship resulted in co-authored publications [J4], [J2], [C3], [C2], [D4], [D3].

Technical Skills

Programming C#, Python, C++.

XR development Unity, Mixed Reality Toolkit (MRTK), Photon networking, Meta Quest 3, HTC Vive.

Methods Computational geometry and optimization, LLM/diffusion-based scene synthesis, quantitative and qualitative HCI evaluation.

Languages

Korean (native); English (fluent).

References

Sung-Hee Lee, Professor, Graduate School of Culture Technology, KAIST, sunghee.lee@kaist.ac.kr (Ph.D. advisor).

Tae-Kyun Kim, Professor, School of Computing, KAIST, kimtaekyun@kaist.ac.kr (Postdoctoral host).